## 2.5 Step 3: Landslide hazard assessment

### 2.5.1 Objective

The objective of this step is to assess of landslide occurrence (and associated magnitude) within a certain time frame and area. As explained earlier this assessment is carried out for the initiation of the shallow landslides alone. The hazard is therefore estimated for the landslide source area.

### 2.5.2 Method

We linked the landslide susceptibility (Step 2) to landslide hazard by assessing the average hazard in different sub regions that are delineated according to their susceptibility (Figure 6). The first sub region encompasses all areas with a landslide susceptibility values between 0 and 0.1, the second sub region all areas with an LSS value between 0.1 and 0.2, and so on. Concretely, for each susceptibility class, the total affected area by the landslide sources (Figure 3 b) that have occurred over the whole period of observation (about 20 years – see Step 1) is averaged yearly. The resulting value provides a landslide affected area in m² year-1 km-2 , i.e. a landslide rate. The combination of a susceptibility (where a landslide occur), with a rate (how often and how strong) characterizes the hazard (Guzzetti et al., 1999; Glade et al., 2006). More details on the assessment of the hazard can be found in Depicker et al. (in review).

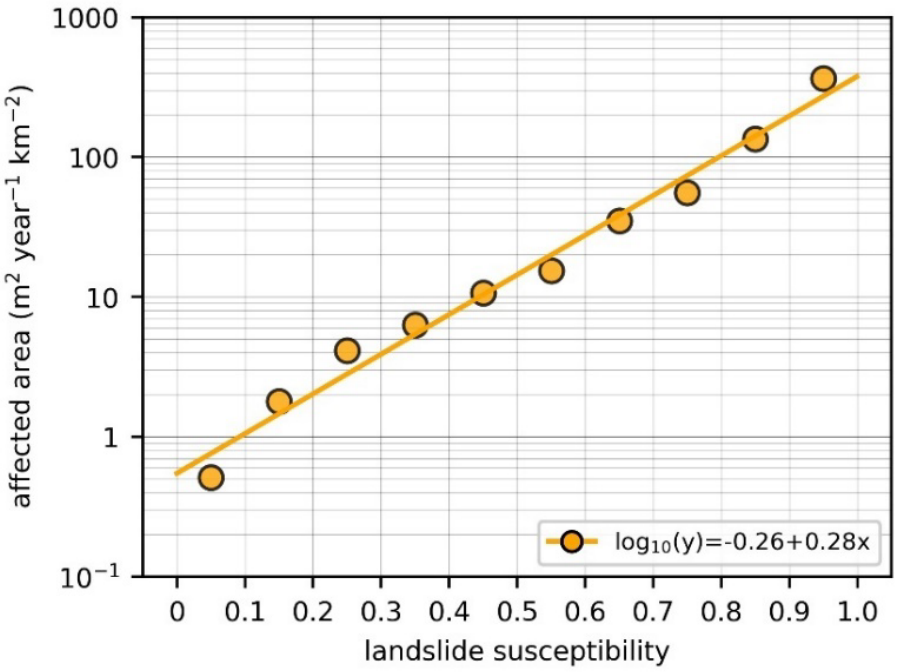



***Figure 6:*** *Relationship between landslide susceptibility and landslide affected area (i.e. landslide rate)*